



IILM UNIVERSITY

Greater Noida, Uttar Pradesh

AN INTERNSHIP REPORT

Artificial Intelligence Internship Programme

Sentiment Analysis Model using Natural Language Processing

STUDENT NAME

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ENROLLMENT NUMBER

2410030383

PROGRAMME & SEMESTER

B.Tech, CSE — 5th Semester

ORGANISATION

Codec Technologies Pvt. Ltd.

DURATION

22/07/2026 – 22/08/2026 (1 Month)

BATCH

2024–2028

Agenda

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About the Organisation

Codec Technologies Pvt. Ltd. profile

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Internship Snapshot

Role, duration, mode & reporting

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Takeaways from the internship

SECTION 1

About Codec Technologies Pvt. Ltd.

A global platform dedicated to empowering learners and connecting diverse talent with meaningful career opportunities worldwide — delivering dedicated IT and business consultancy services across 27+ countries.

- Registered internship provider on the National Internship Portal (AICTE), Ministry of Education, Govt. of India
- Recognised Google for Education Partner
- ISO 9001:2015 certified organisation
- Headquartered at Chandivali IT Hub, Mumbai — hybrid, pan-India delivery model

27+

Countries Served

ISO 9001:2015

Certified

1

Month AICTE-Approved Programme

www.codectechnologies.in | support@codectechnologies.in

Internship Snapshot



Duration

22/07/2026 – 22/08/2026 (1 Month)



Designation

Artificial Intelligence Intern



Location & Mode

Pan India, Hybrid



Reporting To

Assigned Project Head(s)



Programme

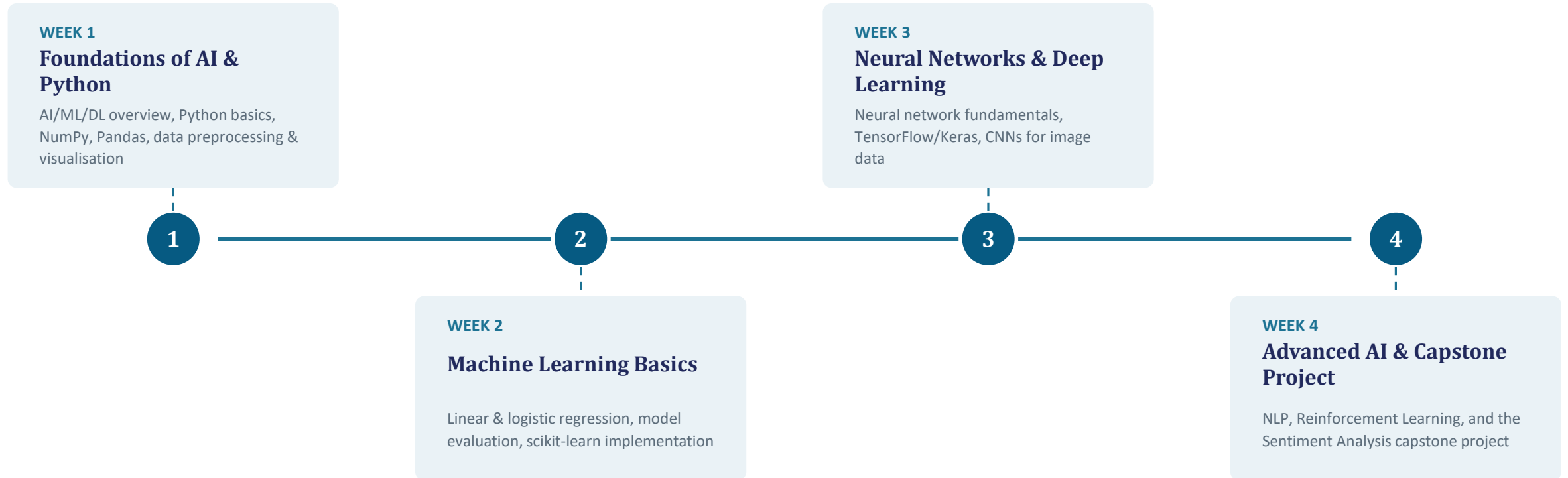
AICTE & ICAC Approved



Completion

21/21 Lessons — 100%

Four-Week Internship Curriculum





Problem Statement

The explosive growth of user-generated text on social media, e-commerce platforms, and review sites means organisations receive enormous volumes of unstructured feedback every day. Manually reading and categorising this feedback is time-consuming, inconsistent, and does not scale — creating the need for an automated system that can accurately classify sentiment as positive, negative, or neutral.

Project Objectives



Understand and apply core NLP concepts — tokenisation, text cleaning, normalisation



Preprocess raw textual data into a numerical form suitable for ML models



Build and train a classification model to predict text sentiment



Evaluate performance using accuracy, precision, recall, and F1-score



Gain hands-on experience with NumPy, Pandas, Scikit-learn, NLTK/SpaCy



Document and present the completed project as the capstone deliverable

Technologies & Tools Used

Core & Data Handling

- Python 3
- NumPy
- Pandas
- Jupyter / Colab

Visualisation

- Matplotlib
- Seaborn

NLP

- NLTK
- SpaCy

Machine Learning

- Scikit-learn
- TF-IDF Vectorizer
- Logistic Regression

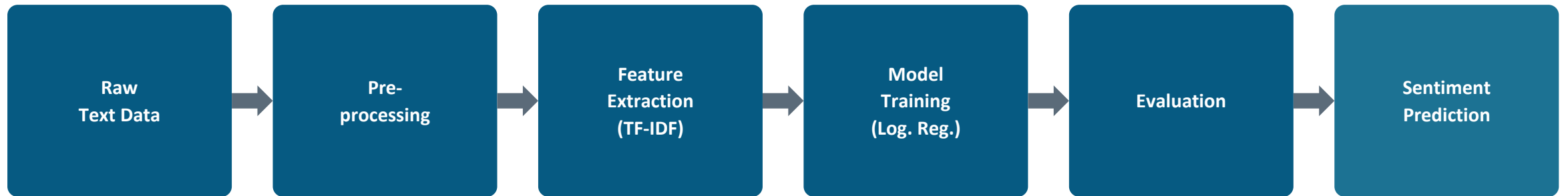
Deep Learning (extension)

- TensorFlow
- Keras

Documentation

- Git
- Markdown

System Architecture



Pipeline: raw text is cleaned and normalised, converted into numerical features via TF-IDF, then classified by a trained Logistic Regression model and evaluated on held-out data before producing a final sentiment label.

Methodology

- 1 Data Collection — labelled text dataset with sentiment labels
- 2 Data Cleaning — handle missing values, duplicates, normalise text
- 3 Exploratory Data Analysis — class distribution & word frequencies
- 4 Text Preprocessing — tokenisation, stop-word removal, lemmatisation
- 5 Feature Engineering — TF-IDF vectorisation
- 6 Train-Test Split
- 7 Model Building — Logistic Regression (Scikit-learn)
- 8 Model Evaluation — accuracy, precision, recall, F1, confusion matrix
- 9 Hyperparameter Tuning — cross-validation
- 10 Documentation & Presentation — final capstone submission

Expected Outcomes

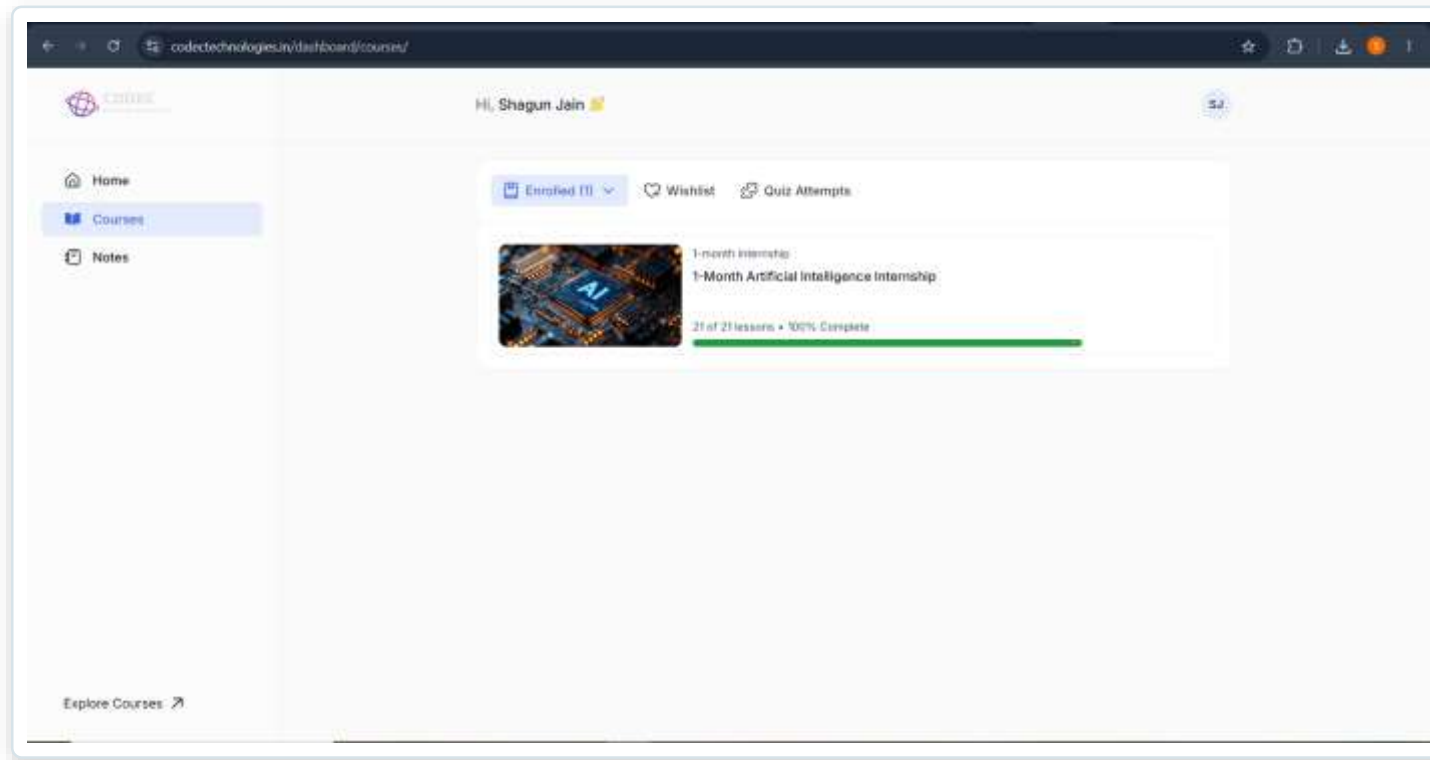
- ✓ A trained sentiment analysis model classifying text into positive, negative, or neutral categories
- ✓ Clear understanding of the end-to-end NLP and ML pipeline, from raw data to a working model
- ✓ Practical proficiency in Python libraries for data handling, visualisation, NLP, and ML
- ✓ Improved ability to evaluate and interpret model performance with standard metrics
- ✓ A completed capstone project consolidating all four weeks of internship learning
- ✓ Successful completion of the internship, reflected in the award of the Certificate of Internship

Certificate of Internship



Awarded to Shagun Jain for successfully completing the 1-Month AICTE & ICAC approved Artificial Intelligence Internship (22/07/2026 – 22/08/2026), issued by Codec Technologies Pvt. Ltd.

Completion & Communication Proof



Codec Technologies learning dashboard — 21 of 21 lessons completed (100%)

All assigned projects and training tasks were completed and submitted on the Codec Technologies platform and via email communication with the assigned project head(s), in accordance with the terms of the internship offer letter.

100%

Course Completion — 21/21 Lessons

Key Learnings & Takeaways

1

Technical Skills

Hands-on proficiency in Python, NLP libraries, and Scikit-learn for building real ML pipelines

2

End-to-End Thinking

Experience across the full pipeline — from raw data to a deployed, evaluated model

3

Industry Exposure

Weekly performance reviews and mentorship simulating a real project environment

4

Professional Growth

Improved documentation, presentation, and independent problem-solving skills

Thank You

Artificial Intelligence Internship | Codec Technologies Pvt. Ltd.

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